Ember Isle: Technical Sheet

Tech used

One self-contained HTML file (about 37 KB) with no external assets or libraries. Rendering uses HTML5 Canvas 2D, DPR-aware (up to 2x), and works in portrait or landscape. All sound effects and both music modes are generated in the browser with the Web Audio API. Audio starts on the first tap, as iOS requires, and is skipped if the browser has no audio support. Best night and wins are saved in localStorage. The simulation core has no DOM code, so it runs headless in Node for testing.

Game systems

- **World:** seeded procedural island (radius 1080) with about 230 trees, 110 rocks, 60 berry bushes and 26 iron veins, plus wandering rabbits. Resources regrow on timers. The ground is pre-rendered once to an offscreen canvas.
- **Day/night:** a 150-second cycle with dusk and dawn fades. Darkness is a half-resolution light map with soft holes cut for fires, torch and shade glows.
- **Survival:** HP, food and warmth with cross-effects (hunger slows you, cold and hunger hurt, being fed and warm heals).
- **Crafting and building:** 10 recipes covering tools, structures, a torch and armor, with placement facing the player and collision for walls and fires.
- **Enemies:** shades steer toward the player and are pushed away by firelight, take fire damage inside it, and hit walls that block them. Brutes ignore light avoidance and deal double damage to walls. Spawn count, health, speed and damage scale with each night.
- **Mobile UX:** floating left joystick, a context-aware ACT button that can be held, large buttons sized to 15% of the short screen edge, auto-pause when the app is backgrounded, and a pause-while-crafting menu.
- **Effects:** layered flickering fires with sparks, additive glows, swaying trees, hit flashes, knockback, screenshake, floating text, and off-screen enemy arrows.
- **Music:** a calm major-pentatonic arpeggio by day and a darker minor pulse at night whose tempo rises each night and adds a bass growl when many shades are close.

Audit 1: Balance (simulated play)

A scripted survivor bot (gathers by need, crafts tools, builds a fire and walls at dusk, refuels, eats and fights) played full runs headless.

Pass	Runs	Avg nights	Shortest	Wins (10 nights)
v1	18	6.5	3	3 (17%)
Final	16	7.3	3	5 (31%)

Fix: night-1 to night-3 deaths were too common, so shade damage changed from 6 + 0.6 per night to 4 + 0.8 per night (gentler start, same late-game threat) and the shade cap dropped by 1. Result: deaths are spread from night 3 to night 8, a skilled player can escape, and every loss in the final pass came from shades rather than hunger or cold, so food and warmth create pressure without being the main killer.

Audit 2: Bugs

The script passed node --check. A Node harness with a stubbed DOM and canvas played through day 1, dusk, a full night and dawn with a scripted thumb player (moving, chopping, eating, opening the craft menu and building) at 60 fps, with 0 runtime errors. Fixed: structures placed at the shoreline could land in the sea, so placement is now clamped inside the island. Inputs and the joystick are cleared when the window loses focus. The game auto-pauses when the app goes to the background.

Audit 3: Player experience

Added: a "Night is coming: build a Campfire!" warning 22 seconds before dark if you have no fire nearby, starving and freezing warnings, a LOW FUEL / OUT label on fires, a gold dot on CRAFT when something is affordable, an ACT label that names the action, a dashed target ring, off-screen shade arrows, a day/night clock with a dusk marker, and first-time control hints.

Caveat: tested with simulated play, not on a physical phone.